



CNN Based Deep Learning Modeling with Explainability Analysis for Detecting Fraudulent Blockchain Transactions

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ABSTRACT

In the era of growing cryptocurrency adoption, Blockchain has emerged as a leading player in the digital payment landscape. However, this widespread popularity also brings forth various security challenges, including the need to safeguard against fraudulent activities. One of the paramount challenges in this regard is the detection of fraudulent transactions within the realm of Bitcoin data. This task significantly influences the trust and security of digital payments. Yet, it's a formidable challenge given the relatively low occurrence of fraudulent Bitcoin transactions. While deep learning techniques have demonstrated their prowess in fraud detection, there remains a scarcity of studies exploring their potential, particularly in Blockchain. This study aims to fill that gap, focusing on our 1D Convolutional Neural Network (CNN) model, which combines the power of eXplainable Artificial Intelligence (XAI) techniques. To understand how our model works and explain its decisions, we use the Shapley Additive exPlanation (SHAP) method, which measures each feature's impact. We also deal with data imbalance by exploring various methods to balance fraudulent and benign Bitcoin transaction data. Our findings are significant, indicating that the proposed 1D CNN model achieves higher accuracy while simultaneously reducing the False Positive Rate (FPR).

1. Introduction

Blockchain technology has gained significant attention due to its decentralized, immutable, and transparent ledger system, which is inherently resistant to data tampering [1]. Originally designed for Bitcoin [2], blockchain's core features—decentralized consensus, cryptographic security, and public verifiability—make it uniquely vulnerable to specific fraud types (e.g., double-spending, Sybil attacks, or wash trading) that traditional financial systems do not face [3]. These fraud patterns necessitate specialized detection methods, as blockchain's pseudonymity and irreversibility amplify the risks of undetected fraudulent transactions. While blockchain's transparency allows transaction tracing, its lack of centralized authority complicates real-time fraud intervention.

Fraud detection in blockchain transactions demands higher precision than conventional systems due to the irreversible nature of transactions. It is crucial to take swift and appropriate measures to minimize potential risks by integrating a real-time fraud detection system, as depicted

in Figure 1. False positives (legitimate transactions flagged as fraud) can disrupt user trust, while false negatives (missed fraud) lead to permanent losses. Our study focuses on Bitcoin transactions because (1) their pseudonymous nature creates distinct fraud detection challenges compared to traditional systems, and (2) their public ledger enables data-driven deep learning approaches while preserving privacy [4]. This contrast underscores the need for explainable AI (XAI) in blockchain fraud detection—a gap our study addresses by combining 1D CNN with SHAP analysis.

However, there is a significant challenge posed by the inherent imbalance between benign and fraudulent data in blockchain datasets, such as those containing Bitcoin or Ethereum transactions. Often, the frequency of fraudulent or anomalous data points is much lower than that of benign or normal data points, resulting in imbalanced datasets. This skewed data distribution can negatively impact the effectiveness of fraud detection systems, which tend to be biased towards the majority class (benign transactions) and struggle to accurately identify the minority class (fraudulent transactions) [5].

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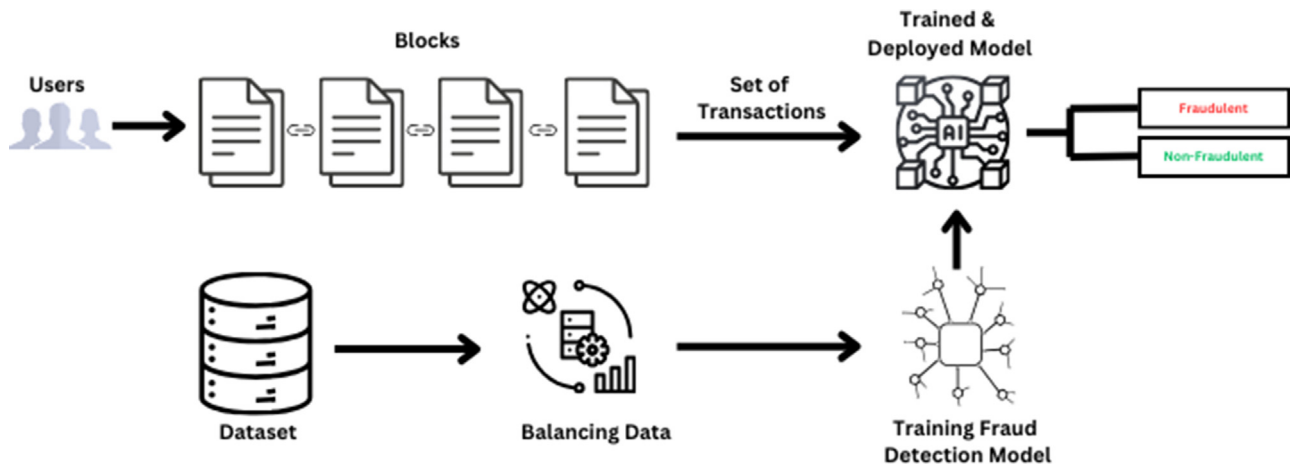


Fig. 1. A System Architecture of the Blockchain Fraud Detection System.

Although many previous studies have improved the accuracy of fraud detection in Blockchain transactions, they often struggle with a high False Positive Rate (FPR) i.e. many legitimate transactions are mistakenly labelled as fraudulent. However, focusing only on accuracy is not enough to assess the performance of a classifier, as high accuracy can exist alongside a low True Positive Rate (TPR) and high FPR. Unlike accuracy, the main goal of a fraud detection system is to identify both fraudulent and benign transactions correctly. Since the dataset contains significantly fewer fraudulent Bitcoin transactions compared to benign transactions, the classifiers are biased towards the majority class (benign transactions). Various techniques like over and under-sampling have been used to tackle these imbalanced data challenges in different domains. Despite the availability of several under-sampling algorithms, each method has its limitations, with one common challenge being the potential exclusion of significant instances that have a major impact on model training.

Additionally, issues such as overfitting, sensitivity to noise, and the introduction of bias into the dataset can reduce the performance of machine learning models [6]. Once the data is balanced, the next hurdle lies in selecting an appropriate classifier for anomaly detection, a task that presents its own set of challenges. Although many studies have opted for tree-based machine learning classifiers for the classification of fraudulent activities [7], in recent times, Deep Learning (DL) algorithms have emerged as a promising approach for addressing the anomaly detection challenge [8,9]. Deep Learning, a subset of Machine Learning (ML), has gained prominence in predictive analysis [10]. In recent times, the 1D Convolutional Neural Network (1D CNN) has garnered attention for its swift feature extraction capabilities, making it a preferred choice for anomaly detection [11], intrusion detection [12], and fault detection [13] across various domains. Despite its proven efficacy in these applications, there is a notable scarcity of studies harnessing the potential of the 1D CNN model for fraud detection within Blockchain transaction data. Recognizing this research gap, our study aims to bridge this void by investigating the effectiveness of the 1D CNN model in the realm of fraud detection in Blockchain transactions.

Although several studies have been conducted in the field, a limitation persists: the lack of explanations for the model's predictions. A fundamental question arises: "Can we have confidence in the predictions made by the Black-Box model?" Explainable Artificial Intelligence (XAI) emerges as a relevant field aiming to address this question. XAI focuses on enhancing the interpretability and transparency of black-box AI models, enabling them to provide comprehensible explanations for complex decisions [14]. Researchers have explored two prominent XAI techniques, namely 'Local Interpretable Model-Agnostic Explanations' (LIME) [15] and 'SHapely Additive exPlanations' (SHAP) [16], to demonstrate the explainability and transparency of AI models. The core

idea behind SHAP is that the influence of a feature is not solely dependent on that individual feature but is interconnected with the entire set of features within the dataset. Therefore, SHAP computes the impact of each feature on the target variable, referred to as the SHAP value.

In this study, we propose a 1D Convolutional Neural Network (1D CNN) model and compare it with widely used tree-based Machine Learning (ML) algorithms to detect fraudulent Bitcoin transactions. We employ tree-based ML classifiers, including Extreme Gradient Boosting (XGB), Random Forest (RF), Decision Tree (DT), Gradient Boosting (GB), and Adaptive Boosting (AdB), for classifying benign and fraudulent Bitcoin transactions. Acknowledging the limitations of undersampling techniques, we explore both oversampling (SMOTE, ADASYN) and combined sampling (SMOTEENN, SMOTETOMEK) methods to address the imbalance in our dataset. Specifically, we applied these techniques exclusively to the training set and evaluated their performance. A comprehensive comparative analysis was conducted among the ML classifiers and the proposed 1D CNN model using over-sampled and combined-sampled data to determine the most suitable sampling technique for identifying fraudulent transactions. Additionally, we validate the prediction accuracy of both the Black-Box ML models and the Deep Learning model through the Explainable Artificial Intelligence (XAI) technique, employing SHAP analysis. This study's contributions are summarized as follows.

- We propose a 1D CNN model designed to identify fraudulent Bitcoin transactions and assess its performance.
- We investigate various over-sampling and combined sampling techniques to address the significant class imbalance in Bitcoin transaction data, offering insights into effective balancing methods through a comprehensive comparative analysis.
- We enhance the interpretability of both CNN and ML classifier predictions by employing SHAP (an eXplainable Artificial Intelligence technique) to identify the key features influencing Bitcoin transaction classification.

The remaining sections of the paper are organized as follows: [Section 2](#) offers a comprehensive overview of recent research, encompassing both machine learning and deep learning techniques. [Section 3](#) delves into the detailed description of the proposed methods. Comparative results are meticulously analyzed and presented in [Section 4](#). A thorough discussion is undertaken in [Section 5](#), while the paper concludes with [Section 6](#).

2. Related Studies

While existing studies have leveraged blockchain intelligence—integrating AI, ML, and DL techniques—to detect fraudulent

transactions [17], critical gaps remain. First, most research has prioritized supervised and unsupervised algorithms, with limited exploration of CNN-based deep learning methods combined with data balancing techniques. Second, the common issue of class imbalance in blockchain transaction datasets undermines model performance, yet only a few of studies systematically evaluate the efficacy of undersampling and oversampling methods to mitigate this challenge.

Aligned with this study, our previous study [18] introduced an ensemble-based anomaly (In this study, we call it fraud) detection system with explainability analysis. Performance metrics such as TPR, FPR, and accuracy were utilized to assess the effectiveness of our ensemble model. Additionally, we explored various tree-based machine learning classifiers, leveraging their faster training capabilities [19], computational efficiency, and ability to handle data without requiring conversion or normalization, particularly in the context of highly imbalanced datasets. The proposed ensemble method demonstrated effectiveness in enhancing accuracy, TPR, and FPR.

For identifying illicit Bitcoin transactions, the study [20] presents a GCN (Graph Convolution Network) model, achieving high accuracy (98.5%) and AUC scores on the Elliptic dataset. The model leverages graph centrality measures and temporal transaction data for fraud detection. Key limitations include limiting direct applicability to other blockchains and relying on labeled data, which may be sparse or noisy. The model's complexity and computational demands could also hinder real-time deployment in large-scale transaction networks.

Another study [21] introduces a novel approach to identifying illicit Bitcoin transactions through a graph neural network (GNN) model that leverages spectral graph convolutions and extracts 19 transaction features. Utilizing a dataset comprising over 13 million transactions from 2,059 entities across 28 user categories, the study conducts both binary (legal vs. illegal) and multi-class (entity classification) experiments. The proposed model achieves up to 92% accuracy, outperforming baseline models like Graph Attention Networks (GAT2) and XGBoost. However, the dataset's reliance on labeled entities may not encompass the full spectrum of illicit behaviors, potentially affecting the model's generalizability.

A machine learning-based fraud detection system for Bitcoin transactions, designed for smart city applications, has been proposed by N. Nayyer et al. [22]. The framework employs an ensemble stacking model, combining multiple classifiers (e.g., Random Forest, XGBoost, and Neural Networks) to improve detection accuracy. It leverages transaction features such as temporal patterns, network topology, and behavioral anomalies to identify fraudulent activities. The model is tested on real-world Bitcoin transaction data, demonstrating superior performance over single-classifier approaches in terms of precision and recall. However, limitations include reliance on labeled data, potential computational overhead from ensemble methods, and challenges in adapting the model to evolving fraud tactics in dynamic blockchain environments.

In addition, the authors in [23] employed five distinct supervised machine learning classifiers, namely Random Forest, Adaptive Boosting, MLP, SVM, and KNN, to identify instances of Bitcoin transaction theft. Furthermore, the authors asserted that their top-performing model surpassed the performance of other unsupervised algorithms, specifically Mahalanobis Distance, LOF, and OCSVM. In their paper [24], the authors employ multiple Machine Learning models to identify anomalous transactions in various digital currency markets. However, neither studies didn't improve the False Positive Rate (FPR), although they had achieved a high accuracy.

Furthermore, The article presents a novel machine learning-based approach called Value-driven-Transactional tracking Analytics for Crypto compliance (VTAC) [25] for detecting illegal cryptocurrency transactions on blockchain networks. VTAC combines machine learning algorithms (Random Forest, ADA Boost, and XG Boost) with pre-training, normalization, and de-anonymization processes to analyze transactional data, achieving a detection accuracy of 97.5%. In [26], the authors have conducted a comparative analysis of Bitcoin and Ethereum

transaction data using various under and oversampling techniques. In a separate study detailed in [27], the authors have compared Light Gradient Boosting Machine (LGBM) with other machine learning models (e.g., Random Forest, XGBoost, MLP) on a dataset of 9,841 transactions, highlighting LGBM's superior performance in terms of speed, memory efficiency, and accuracy. However, limitations include feature selection and scalability enhancements.

In contrast, a hybrid deep learning approach is proposed by Q Umer et al. [28], combining Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to detect fraudulent transactions in cryptocurrencies. Key strengths include its ability to capture both spatial (via CNN) and temporal (via LSTM) dependencies in transaction data. However, limitations include high computational complexity, reliance on large labeled datasets (which are scarce for fraud cases), and potential challenges in generalizing across different blockchain networks due to varying transaction structures.

Moreover, in [29], the authors introduced a novel approach for Bitcoin transaction analysis to detect potential money laundering anomalies. It leverages a combined method, integrating random forests with information from a graph convolutional network. The model's results suggest the presence of possible shadow transactions, estimated at 2-3% of the total market. The authors in [30] present a deep learning-based anomaly detection model in the financial sector, employing Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and one-dimensional Convolutional Neural Network (1dCNN) algorithms.

Additionally, a one-dimensional CNN model for network anomaly detection in cybersecurity is introduced by the authors in [31]. It categorized network traffic data into TCP, UDP, and OTHER protocols, addressing class imbalance through over-sampling after feature selection using the Chi-square technique. In [32], the authors have devised an encoder-decoder-based deep learning model to detect anomalous activities within the Ethereum transaction network. On the other hand, the authors in [33] employ a semi-supervised GAN (SGAN) framework to identify fraudulent Ethereum transactions by leveraging both labeled and unlabeled data. They were able to improve detection accuracy in imbalanced datasets, extract effective features from transaction graphs and temporal patterns, and outperform traditional supervised methods like Random Forest and SVM in experiments. However, limitations involve high computational costs due to GAN training complexity.

While previous studies focused on employing supervised and unsupervised algorithms solely for classifying fraudulent transactions in blockchain transaction data, limited attention has been given to addressing the tradeoff between TPR and FPR. In addition, Class imbalance poses challenges in detecting anomalies in areas such as illicit activities in digital currencies, money laundering, and various single or multi-class classifications [34]. To tackle this issue, researchers have introduced various undersampling and oversampling methods aimed at enhancing evaluation metrics. In studies [35] to [36], SMOTE oversampling balancing methods have been applied to classify illegal activities in Ethereum Blockchain transactions while the authors of [36] and [37] have explored various under-sampling techniques. However, very few studies have utilized the effectiveness of CNN-based DL methods for detecting fraudulent activities in Blockchain transactions. Additionally, A key limitation is the absence of explanations for model predictions.

3. Methodology

The overall methodology for this work is visually represented in Figure 2. The initial phase involved dataset collection and subsequent preprocessing. Following this, the dataset was split into training and test data partitions. Notably, data sampling was exclusively applied to the training data, while the test data remained independent. Initially, both ML and CNN models were trained without balancing the data to observe the impact of data imbalance. Also, the over-sampled data was utilized to train both machine learning classifiers and the CNN model. Ultimately, independent test data was employed to validate the mod-

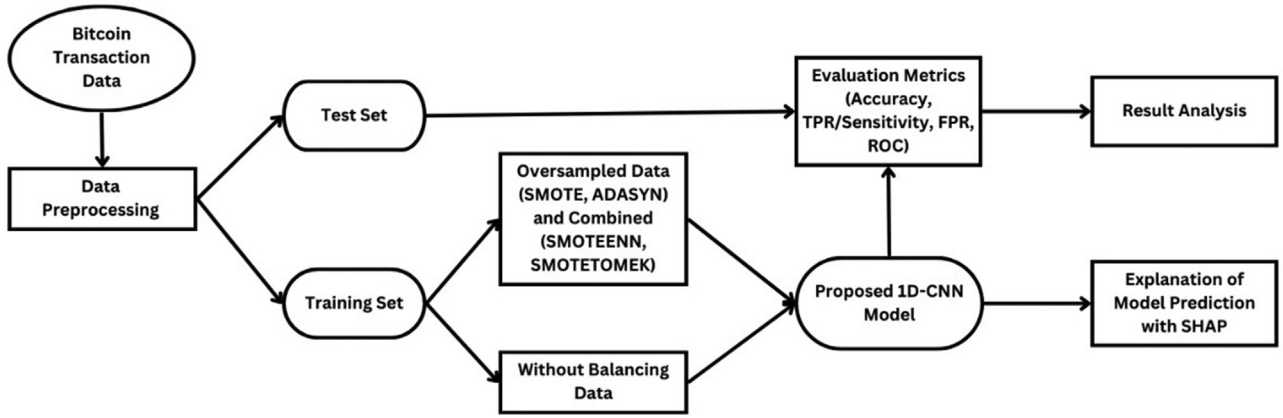


Fig. 2. Methodology of this work.

els, utilizing a range of evaluation metrics including Accuracy, TPR or Sensitivity, FPR, and ROC-AUC score. Furthermore, the study incorporates a comparative analysis, complemented by eXplainable Artificial Intelligence (XAI) techniques, which are elaborated upon in the results section.

3.1. Dataset

The Bitcoin transaction data utilized in this study was sourced from the IEEE Data Portal¹. This dataset comprises a total of 30,248,134 samples, with the majority, specifically 30,248,026, being labelled as negative samples, denoting benign transactions. In stark contrast, there are only 108 samples labelled as fraudulent. This dataset exhibits a significant class imbalance, emphasizing the scarcity of fraudulent instances. Exploratory Data Analysis (EDA) was conducted to gain insights into the dataset's characteristics. It encompasses 12 attributes, with each transaction being assigned a label indicating its status as fraudulent (labelled as 1) or benign (labelled as 0).

Features such as *in_btc*, *out_btc*, *total_btc*, *mean_in_btc*, and *mean_out_btc* exhibit strong positive correlations. Conversely, the *indegree* and *outdegree* features display weak correlations. While *in_malicious*, *out_malicious*, *is_malicious*, and *all_malicious* features are notably correlated with the output feature *out_and_tx_malicious*, no substantial correlations are observed between these features and *indegree*, *outdegree*, *in_btc*, *out_btc*, *total_btc*, *mean_in_btc*, and *mean_out_btc*.

Furthermore, a hypothesis test for feature selection was conducted using the T-test to explore the correlations between positive (Fraudulent Bitcoin transactions) and negative (benign Bitcoin transactions) samples. The T-test assesses whether a significant difference exists between the means of the positive and negative samples. T-statistic values and corresponding p-values were computed for each attribute, as presented in Table 1. Notably, all attributes exhibited significance with p-values below 0.01, except for *outdegree*. It's also important to mention that all features (*in_malicious*, *out_malicious*, *is_malicious*, and *all_malicious*) share the same value range as the target feature *out_and_tx_malicious*. Consequently, this similarity poses a challenge for ML classifiers in effectively distinguishing Bitcoin transactions, leading to the exclusion of these four features and *outdegree*. As a result, a set of seven features, including the target feature, was selected for classification. A summary of the selected features is presented in Table 2.

After completing the feature selection process, the dataset was partitioned into negative and positive samples, focusing on eliminating duplicate entries exclusively from the negative samples having the same values for all selected features. To manage computational complexity,

Table 1

The T-values and P-values for all Features.

Feature	T value	P value
<i>indegree</i>	-14.013838	0.000000
<i>outdegree</i>	0.842249	0.399648
<i>in_btc</i>	-17.229753	0.000000
<i>out_btc</i>	-16.469202	0.000000
<i>total_btc</i>	-16.864202	0.000000
<i>mean_in_btc</i>	-8.727102	0.000000
<i>mean_out_btc</i>	-16.014732	0.000000
<i>in_malicious</i>	-68.869826	0.000000
<i>out_malicious</i>	-5432.702805	0.000000
<i>is_malicious</i>	-3878.622465	0.000000
<i>all_malicious</i>	-1866.899584	0.000000

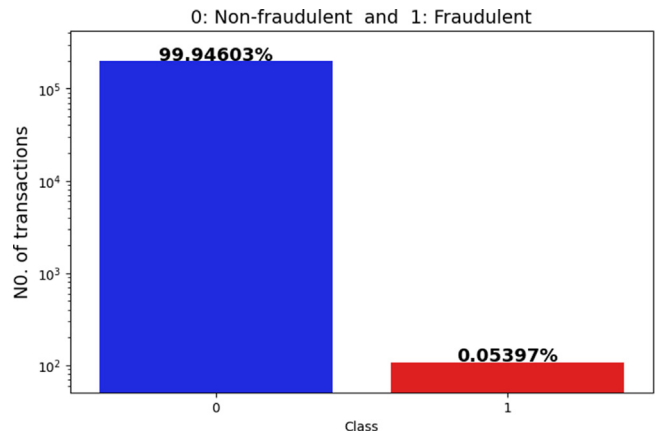


Fig. 3. Frequency of the classes.

a decision was made to retain only 200,000 negative samples, while maintaining the 108 positive samples. To choose the desired samples from the negative instance set, n iterations were performed where n equals the total count of negative samples divided by 2 lacs. Each iteration saved the randomly selected subset of negative samples with the highest accuracy (generally the accuracy varied between .95 and .99) for further processing. However, it's important to note that this choice still resulted in a significantly high imbalance ratio, as depicted in Figure 3.

3.2. Imbalanced Data Handling

In Bitcoin transactions, the frequency of fraudulent transactions is considerably lower compared to that of normal transactions, resulting in an imbalanced dataset. This imbalance can lead classifiers to exhibit

¹ <https://ieee-dataport.org/open-access/bitcoin-transactions-data-2011-2013>

Table 2
Description of the selected features.

Feature Name	description
indegree	No. of inputs for a given transaction
in_btc	No. of Bitcoins on each incoming edge to a given transaction
out_btc	No. of Bitcoins on each outgoing edge from a given transaction
total_btc	Total number of Bitcoins for a given transaction
mean_in_btc	Average number of Bitcoins on each incoming edge to a given transaction
mean_out_btc	Average number of Bitcoins on each outgoing edge from a given transaction
out_and_tx_malicious	Status of a given Bitcoin transaction if it is fraudulent or not

a bias toward the majority class, as discussed in [38]. While classification accuracy may seem acceptable in many instances, there often exists a noticeable disparity between TPR and FPR values. This discrepancy suggests that the models encounter challenges in accurately classifying fraudulent transactions. To mitigate this issue, it is imperative to balance the dataset by employing built-in or customized sampling techniques before training the classification models.

In scenarios related to fraud detection or money laundering, it is common for positive cases to be rare. Consequently, undersampling techniques can be valuable for rebalancing the dataset while emphasizing the accurate identification of positive cases. Nevertheless, in situations where the number of positive cases, or anomalies within the minority class, is exceptionally low, Machine Learning or Deep Learning models may end up being trained on a restricted dataset created through the use of undersampling techniques.

Conversely, over-sampling methods aim to augment the number of instances in the minority class to align it with the majority class. While these methods involve creating synthetic data by combining majority and minority samples, they can be quite effective. In our study, we investigate well-known over-sampling techniques such as SMOTE and ADASYN, along with combined approaches like SMOTEENN and SMOTETOMEK.

3.2.1. Over-Sampling Techniques

In our investigation, we have explored two commonly used over-sampling techniques to address class imbalance in Bitcoin transaction data. Between these two methods, the Synthetic Minority Over-sampling Technique (SMOTE) stands out as a widely adopted approach for mitigating the class imbalance problem [39], particularly in the context of Bitcoin Transactions for fraud detection. The core objective of SMOTE is to rectify the class distribution imbalance by creating synthetic instances of the minority class. This process helps alleviate bias issues and enhances the generalization of the model. For each minority sample X_i , SMOTE randomly selects k nearest neighbors from the same class. Subsequently, a new sample X_n is generated using one of the nearest neighbors X_{zi} from the set of k neighbors. The generation of the new sample follows the equation 1.

$$X_n = X_i + \lambda * (X_{zi} - X_i) \quad (1)$$

Where λ represents a random number ranging from 0 to 1, consequently, a new synthetic instance is created within the feature space. This process iterates until the number of samples in both the majority and minority classes becomes equal. Ultimately, the original minority instances are combined with the newly generated synthetic instances to establish a balanced dataset.

ADASYN [40] (Adaptive Synthetic Sampling) employs the same formula as mentioned earlier for generating new samples, with the exception being the selection of X_i . ADASYN focuses on enhancing the density of synthetic instances in regions that pose greater classification challenges, offering a more nuanced approach to addressing class imbalance. For each minority instance, ADASYN initially calculates the number of its k nearest neighbors that belong to the majority class, indicating the proximity of the minority sample to the majority class. Subsequently, it

calculates the imbalance ratio α_{os} for each minority instance using the equation 2.

$$\alpha_{os} = \frac{N_{rm}}{N_m} \quad (2)$$

Here, N_{rm} represents the number of samples in the minority class after resampling, and N_m denotes the number of samples in the majority class. This imbalance ratio is employed to determine the desired count of synthetic instances to be generated for the current minority instance. Additionally, a difficulty ratio is calculated to assess the level of difficulty. When this ratio is high, more neighbors are considered for generating synthetic instances. By interpolating feature values between the minority instance and its selected neighbors, a new synthetic instance is generated. This process continues for all minority instances until the number of samples in both the majority and minority classes is equalized. Ultimately, the original minority instances are merged with the newly generated synthetic instances to construct a balanced dataset.

3.2.2. Combined-Sampling Techniques

Additionally, we have explored two combined sampling strategies to address class imbalance in Bitcoin transaction data. SMOTEENN [41] is a combination of two resampling techniques: SMOTE (Synthetic Minority Over-sampling Technique) and Edited Nearest Neighbors (ENN). This approach is designed to address the class imbalance by initially generating synthetic samples using SMOTE and subsequently enhancing the dataset quality through the application of ENN. The primary goal of SMOTEENN is to provide a more sophisticated approach to balancing imbalanced datasets while concurrently improving dataset quality by eliminating potential noise. Following the generation of synthetic instances via SMOTE, ENN operates as follows: for each instance in the dataset, ENN identifies its k nearest neighbors. If the instance's class differs from the majority class of its neighbors, ENN removes the instance from the dataset, effectively eliminating noisy or misclassified instances.

Another combined resampling technique is SMOTETOMEK [42], where synthetic instances are produced using SMOTE, and the undersampling technique Tomek Link identifies pairs of instances from different classes that are closest to each other. For each pair of instances identified as Tomek links, the majority class instance is removed. This undersampling method assists in eliminating instances that are near the decision boundary and may be prone to misclassification. The outcome is a more balanced, discriminative, and effective dataset for training machine learning models.

3.3. ML Models

A Decision Tree classifier is a popular machine learning algorithm that builds a tree-like structure of decisions based on features to make predictions [43]. The decision tree starts with a root node and splits the data into subsets based on feature values to create child nodes. Each node represents a decision or test on a feature such as *indegree*, *in_btc*, *out_btc*, *total_btc*, *mean_in_btc*, and *mean_out_btc*. Decision Trees use Gini Impurity and Information Gain (Entropy) to determine the best feature

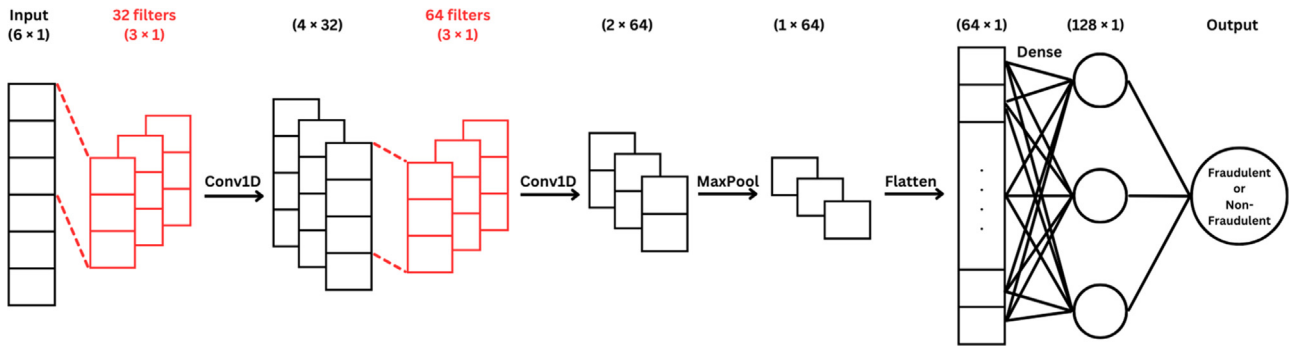


Fig. 4. Architecture of proposed 1D CNN model.

and value to split the data at each node. Moreover, the Random Forest classifier, popular for its ability to reduce overfitting, takes the ensemble of decision trees, each trained on a different subset of data and features, collectively improving the model's predictive performance and generalization [44].

On the other side, Gradient Boosting [45], Extreme Gradient Boosting (XGBoost) [46], and AdaBoost [47] classifiers are ensemble Machine Learning techniques used for both classification and regression tasks. These ensemble methods are powerful and often achieve high predictive accuracy. Generally, the tree-based classifiers combine the predictions of multiple weak learners, typically decision trees, to create a strong predictive model. AdaBoost combines the predictions of all base learners by weighted majority voting. Gradient Boosting starts with an initial base learner, often a decision tree, to make predictions. The final prediction is obtained by summing up the predictions of all the models, effectively creating a weighted combination of the weak learners' predictions.

3.4. Proposed 1D CNN

The 1D CNN architecture, as shown in Figure 4, is designed to extract and transform features from Bitcoin transaction data through convolutional layers, normalize activations with batch normalization, reduce spatial dimensions with max-pooling, and make a binary classification decision in the output layer. It encompasses various layers and techniques to balance model complexity and performance while preventing overfitting, ultimately serving as a robust tool for detecting fraudulent transactions within Bitcoin data.

The feature extraction process begins with Conv1D layers, two in total, each serving a unique purpose. These layers apply convolutional operations to the input sequences, enabling them to capture intricate patterns and representations. The first Conv1D layer is responsible for extracting important features from the input data ($6, 1$) where 6 represents the length of the input sequence and 1 for the dimension of each feature (usually 1 for scalar values). It uses 32 filters to scan the input data with a kernel size of 3, allowing it to capture local patterns within the input sequence. It is also followed by a Rectified Linear Unit (ReLU) activation function to introduce non-linearity. The second Conv1D layer follows a similar pattern but utilizes 64 filters, thereby enabling the extraction of increasingly complex patterns from the data.

Batch Normalization layers are strategically placed after each Conv1D layer to stabilize and accelerate the training process. These layers normalize the activations produced by the convolutional layers, ensuring they have a mean of 0 and a standard deviation of 1. This normalization is pivotal in improving training stability and convergence.

To reduce the spatial dimensions of the feature maps while retaining essential information, a MaxPooling1D layer is incorporated. This layer effectively halves the sequence length, thus reducing computational complexity and retaining the most significant information present in the data. The flattened layer follows the MaxPooling1D layer, reshaping the 3D tensor output into a 1D vector. This transformation prepares

the data for processing by the fully connected layers, ensuring compatibility between the convolutional and dense layers.

The architecture employs two dense (fully connected) layers for classification. The first dense layer consists of 128 neurons and incorporates the ReLU activation function. Additionally, it applies dropout regularization with a 50% dropout rate, which helps mitigate overfitting. The second dense layer functions as the output layer, housing a single neuron, and uses the sigmoid activation function for binary classification. This layer is responsible for making the final prediction regarding the nature of the Bitcoin transaction. Details of the Layers of the Proposed model are given in Table 3.

For optimization, we employed the Adam stochastic gradient descent method with a learning rate of 0.0001. During training, binary cross-entropy was utilized to compute the loss. A batch size of 32 was employed, implying that 32 samples were used per gradient update. A list of parameters used for model compilation is given in Table 4. Given the approximately 320,000 samples in the training set, this resulted in $(320,000 \div 32) = 10,000$ (approximately) steps needed per epoch to iterate over the entire training set. Initially, we used 100 epochs for training our proposed CNN model and monitored the accuracy and loss updates for both the training and validation sets. The number of epochs was determined using the early stopping technique with the patience of 10 epochs to restore the model's best weights to mitigate overfitting.

4. Results Analysis

In this section, we evaluate the proposed 1D CNN model, considering both oversampled and combined sampled data. Furthermore, we present a comprehensive comparative analysis between the performance of machine learning classifiers and the proposed 1D CNN model. We commence by establishing the experimental environment and subsequently provide detailed insights into various aspects of the model's performance concerning Bitcoin fraud detection.

4.1. Environment Setup

We employed the Python programming language for implementation and utilized Jupyter Notebook to execute all modules. The computational tasks benefited from the capabilities of Google Colab, which leveraged 12.68 GB of RAM and 225.83 GB of disk space for enhanced hardware support. A suite of libraries and packages played integral roles in diverse tasks: Pandas facilitated data extraction, Matplotlib enabled visualization and plotting, Numpy performed various mathematical functions, Scikit-learn handled tasks such as binarizing output, dataset splitting, and generating classification reports and confusion matrices, Keras was instrumental in building models using the TensorFlow library, Seaborn was employed for plotting confusion matrices, and shap contributed to explaining the model's output.

Table 3
Details of the Layers of the Proposed CNN Model.

Layer	Kernel Size	No. of Filters	Activation	Output Shape	Parameters
Conv1D	3	32	relu	(None, 4, 32)	128
BatchNormalization				(None, 4, 32)	128
Conv1D	3	64	relu	(None, 2, 64)	6,208
BatchNormalization				(None, 2, 64)	256
MaxPooling1D	2			(None, 1, 64)	0
Flatten				(None, 64)	0
Dense			relu	(None, 128)	8,320
Dense			sigmoid	(None, 1)	129
Total Parameters					15,169

Table 4
Parameters for Model Compilation.

Learning Rate	0.0001
Optimizer	Adam
Batch Size	32
Loss Function	binary_crossentropy

Table 5
Comparison among the classifiers without balancing the data.

Classifiers	Accuracy	TPR	AUC-Score
DT	0.99	0.0	0.55
GBoost	0.99	0.09	0.62
XgBoost	0.99	0.0	0.85
RF	0.99	0.0	0.72
AdaBoost	0.99	0.05	0.82
1D CNN	0.99	0.0	0.49

4.2. Evaluation metrics

In assessing the performance of our fraud detection system, we go beyond accuracy and incorporate essential metrics like True Positive Rate (TPR) and False Positive Rate (FPR). These metrics serve to evaluate the accurate identification of fraudulent and benign transactions, aligning with the core objective of our study. To compare the effectiveness of our proposed 1D CNN model, we employ evaluation metrics such as accuracy, TPR, and FPR. Furthermore, we consider the Receiver Operating Characteristic (ROC) score, which juxtaposes TPR against FPR. Below, we provide definitions for these performance metrics.

TP = True Positive: a fraudulent Bitcoin transaction is correctly identified as fraudulent

TN = True Negative: a benign Bitcoin transaction is correctly identified as benign

FP = False Positive: a benign Bitcoin transaction is incorrectly identified as fraudulent

FN = False Negative: a fraudulent Bitcoin transaction is incorrectly identified as benign

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$TPR = Sensitivity = \frac{TP}{TP + FN} \quad (4)$$

$$TNR = Specificity = \frac{TN}{TN + FP} \quad (5)$$

$$FPR = \frac{FP}{TN + FP} \quad (6)$$

AUC Score is calculated as the Area Under the *Sensitivity(TPR)* - (1 - *Specificity(FPR)*) Curve.

4.3. Impact of data balancing in fraud detection with 1D CNN

We split the dataset into 80% for training and 20% for testing. Initially, the classifiers were trained on the unbalanced training set, which introduced a bias toward the majority class. As a result, the classifiers achieved a high true negative rate but struggled to accurately detect fraudulent transactions, often yielding very low or even zero true positive rates.

Table 5 provides a comparison of accuracy, TPR or Sensitivity, and ROC-AUC score for the Decision Tree (DT), Gradient Boosting (GBoost), Extreme Gradient Boosting (XGBoost), Random Forest (RF), Adaptive Boosting (AdaBoost), and 1d CNN classifiers. Notably, the TPR scores

Table 6
Comparison of evaluation metrics between Over and Combined sampling with 1D CNN.

Model	Accuracy	TPR/Sensitivity	FPR	AUC-Score
CNN+SMOTE	0.98	0.36	0.02	0.76
CNN+ADASYN	0.99	0.27	0.005	0.83
CNN+SMOTEENN	0.99	0.27	0.0008	0.76
CNN+SMOTETOMEK	0.98	0.32	0.02	0.78

were zero for DT, XgBoost, RF, and 1D CNN, indicating that no fraudulent transactions were correctly identified; instead, all transactions were classified as benign. Although GBoost and AdaBoost classifiers showed a low TPR, their scores approached zero. This suggests that fraudulent transactions remained undetected due to the models' bias towards the majority class i.e. benign transactions.

As we trained our proposed CNN model on the unbalanced training set, the model displayed a high true negative rate but faced challenges in correctly identifying positive samples i.e. the Bitcoin fraudulent transactions, resulting in zero true positive rate. While keeping the test set independent, we balanced the training set using various sampling techniques, including SMOTE, ADASYN oversampling methods, as well as SMOTEENN and SMOTETOMEK combined sampling techniques. Subsequently, the CNN model was trained on each of the sampled datasets.

Table 6 illustrates that the CNN model's accuracy remains consistent across different sampled datasets. Among these datasets, the CNN model with SMOTE oversampling exhibits the highest True Positive rate (0.36), emphasizing its effectiveness in identifying fraudulent Bitcoin transactions. In contrast, the CNN model with SMOTEENN demonstrates the best FPR score (0.0008), indicating its capability to detect all benign transactions i.e. True Negative rate is almost 100%. Remarkably, the ROC-AUC scores for the CNN model remain consistent across all sampled datasets. These results are achieved after certain epochs based on early stopping. Overall, the effectiveness of the proposed CNN model is evident in reducing the False Positive Rate (FPR) with both over and combined sampling methods, despite the True Positive Rate (TPR) not meeting the desired level.

While early stopping was implemented to mitigate overfitting issues, we additionally conducted a run of the CNN model for 100 epochs to assess potential improvements in the results. Table 7 presents the evaluation metrics, including accuracy, FPR, and TPR, for both

Table 7
Comparison of the results with and without callback.

Model	Early Stopping			100 epochs		
	Accuracy	FPR	TPR	Accuracy	FPR	TPR
CNN+SMOTE	0.98	0.02	0.36	0.95	0.05	0.45
CNN+ADASYN	0.99	0.005	0.27	0.92	0.08	0.45
CNN+SMOTEENN	0.99	0.0008	0.27	0.97	0.03	0.41
CNN+SMOTETOMEK	0.98	0.02	0.32	0.99	0.01	0.32

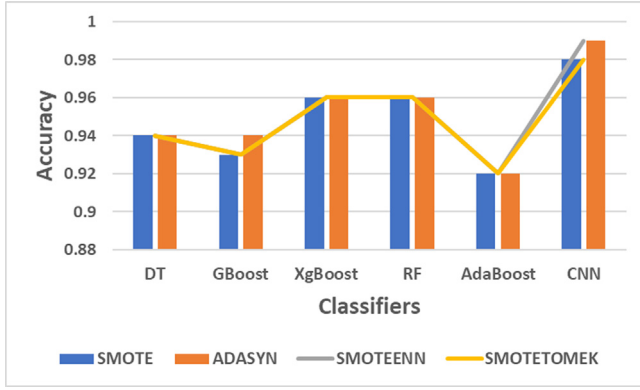


Fig. 5. Comparison of accuracy.

cases. Notably, the TPR or Sensitivity has increased in all models, except for CNN+SMOTETOMEK, which remains constant when trained for 100 epochs. However, there is a significant increase in the False Positive Rate when using 100 epochs, indicating a trade-off between FPR and TPR. The validation accuracy has generally decreased, except CNN+SMOTETOMEK, which exhibits a slight increase after 100 epochs. This behavior suggests that the model oscillates between focusing on benign and fraudulent transactions, leading to increased False Positive (FP) values despite improvements in True Positive (TP) values. The models stopped training at different epochs: 48, 36, 41, and 31 for CNN+SMOTE, CNN+ADASYN, CNN+SMOTEENN, and CNN+SMOTETOMEK, respectively.

4.4. Performance Analysis of the Proposed CNN model

A comprehensive comparative analysis was carried out to evaluate the performance of machine learning (ML) classifiers and the proposed 1D CNN model using over-sampled and combined-sampled data. Figure 5 demonstrates that our 1D CNN model consistently outperforms all tree-based classifiers, achieving an impressive 100% accuracy across all sampled data. In contrast, the tree-based classifiers exhibit accuracies ranging from 0.92 to 0.96. However, figure 6 reveals that the ML classifiers consistently achieve higher True Positive Rates when compared to the 1D CNN model. Notably, among the ML models, Adaptive Boosting stands out with the highest TPR value of 0.59 when using SMOTEENN sampled data.

However, In figure 7, the 1D CNN achieves the best False Positive Rate of 0.0008 with SMOTEENN sampled data, while the ML methods achieve FPR scores ranging from 0.04 to 0.07. Although the CNN model exhibits slightly lower TPR, its robust FPR score leads to the correct identification of most of the benign transactions. In addition, Figure 8 depicts that our proposed 1D CNN model exhibits improved ROC-AUC scores using the ADASYN over-sampling technique, outperforming the scores achieved by the tree-based ML classifiers. This suggests that our proposed CNN model also performs better in terms of ROC-AUC scores. Additionally, We have achieved a remarkable improvement in minimizing the FPR score (0.0008) and a good improvement in increasing the Accuracy (0.99).

The comparative analysis demonstrates the superior performance of the proposed 1D CNN method against existing approaches for detecting fraudulent blockchain transactions, as illustrated in Table 8. While Heidari et al. (GCN & MLP) [48] and Dahal et al. (GNN) [20] achieved high accuracy (0.995 and 0.985, respectively), neither incorporated explainability, and both had same false positive rates (0.01). Nayyer et al. [22] used a stacking ensemble (RF, DT, NB, KNN) with explainability but yielded lower accuracy (0.97) and a higher FPR (0.03). Similarly, using the same baseline models as our, A. Venčkauskas et al. (RF, ADA Boost, XGBoost) [25] reached 0.975 accuracy but lacked explainability and had an FPR of 0.03.

In contrast, the 1D CNN method outperforms all prior works, achieving 0.99 accuracy—nearly matching the highest benchmarks—while drastically reducing the FPR to just 0.0008, the lowest among all compared models. Additionally, it introduces explainability, a critical feature absent in most previous studies. This advancement ensures not only highly precise fraud detection but also interpretability, addressing a key limitation in earlier approaches. The success of the 1D CNN stems from optimized preprocessing, feature selection, and deep learning architecture, making it a robust solution for real-world blockchain transaction monitoring.

4.5. Explainability Analysis using SHAP

SHAP is a potent technique employed to elucidate the predictions of machine learning models. It offers a means to assign the contribution of each feature to a particular prediction, enhancing our ability to explain why a model reached a specific decision. This study leverages the SHAP KernelExplainer method to calculate SHAP values, elucidating the behavior of both ML and DL models for individual instances. We employed the model-agnostic SHAP method to interpret both Machine Learning (ML) and Deep Learning (DL) models.

SHAP values serve the purpose of explaining how a specific feature's value influences the model's prediction compared to what the prediction would be when that feature adopts a baseline value. At first, it establishes a baseline prediction, typically by using the average prediction of the model for a set of instances. This serves as a reference point. For each feature permutation, SHAP predicts the model's output and calculates the difference between the baseline prediction and the current prediction. Then, the method averages these differences across all permutations to determine the SHAP value for the feature being evaluated.

Figure 9 provides a visual comparison of SHAP values, showcasing the contributions of features for four randomly selected fraudulent and benign transactions, as predicted by both the CNN and ML models. In Figure 9, the baseline value for expected fraudulent transactions is 0.27 for the CNN model and 0.50 for the ML models. Remarkably, the predicted scores for fraudulent Bitcoin transactions are 0.89 with the CNN model and 0.86 with the ML models, which are notably higher than the baseline values. The SHAP value representations in Figures 9(a) and 9(b) illuminate the features contributing the most to increasing the score. These figures highlight that "total_btc" holds the highest importance for the CNN model and the second highest for ML models, followed by "in_btc" or "mean_out_btc," "out_btc". On the other hand, with the same baseline value, the predicted scores for benign Bitcoin transactions are 0.03 with the CNN model and 0.40 with the ML models, which

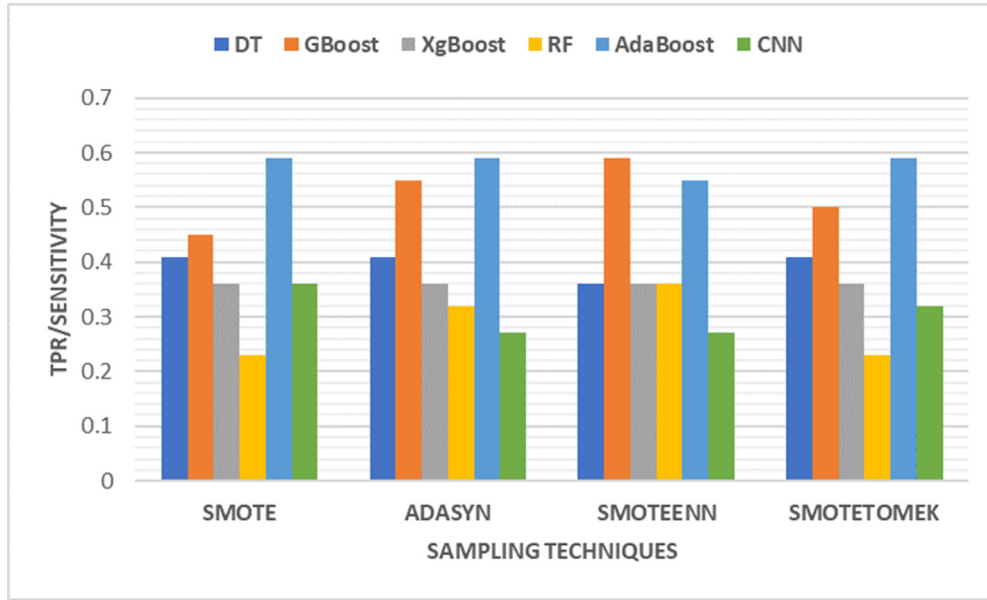


Fig. 6. Comparison of TPR/Sensitivity.

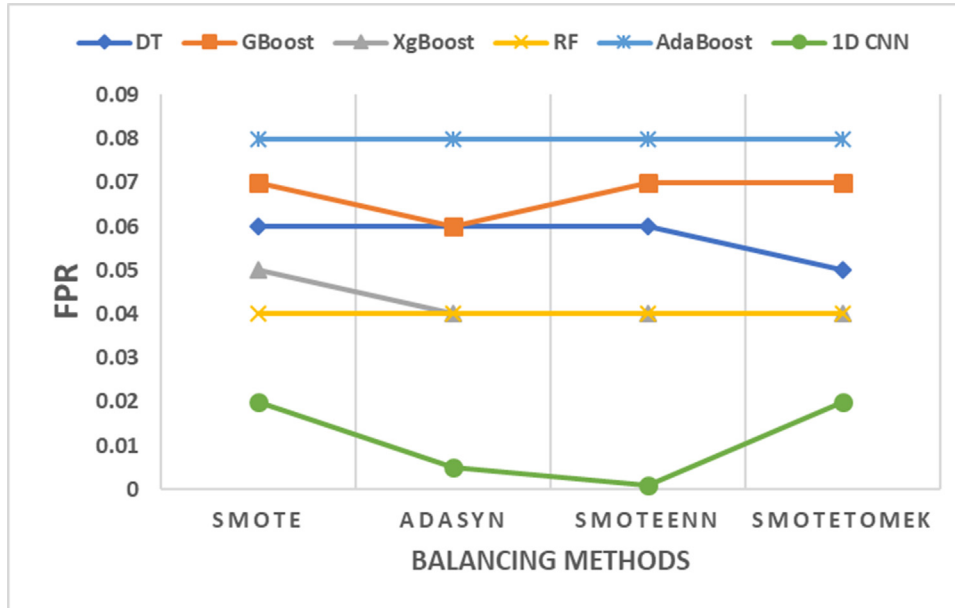


Fig. 7. Comparison of FPR.

Table 8
Comparison between Proposed and Existing Work for detecting fraudulent transactions.

References	Model	Accuracy	FPR	Explainability
Heidari et al.	GCN & MLP	0.995	0.01	No
Nayyer et al.	Stacking Ensemble (RF, DT, NB, KNN)	0.97	0.03	Yes
A. Venčkauskas et al.	RF, ADABOOST, XGBOOST	0.975	0.03	No
Dahal et al.	GNN	0.985	0.01	No
Our Study	1D CNN	0.99	0.0008	Yes

are lower than the baseline values. The SHAP value representations in Figures 9(c) and 9(d) elucidate the features contributing the most to decreasing the score. These figures emphasize that "total_btc" holds the highest importance for the CNN model and the second highest for ML models, followed by "out_btc," or "indegree".

Although "mean_in_btc" and "in_btc" hold the highest positions for ML models in fraudulent and benign cases, respectively, "total_btc" contributes the most in both cases. Comparing all four cases in Figure 9, it's evident that the "total_btc" values for fraudulent instances are higher

than those for normal instances, making them distinguishable to human observation. Hence, it can be concluded that SHAP values are effective in explaining both normal and fraudulent Bitcoin transactions.

5. Discussion

Our primary objective is to improve the detection accuracy of both fraudulent and benign transactions in highly imbalanced Blockchain datasets. Detecting fraudulent transactions under such imbalance is

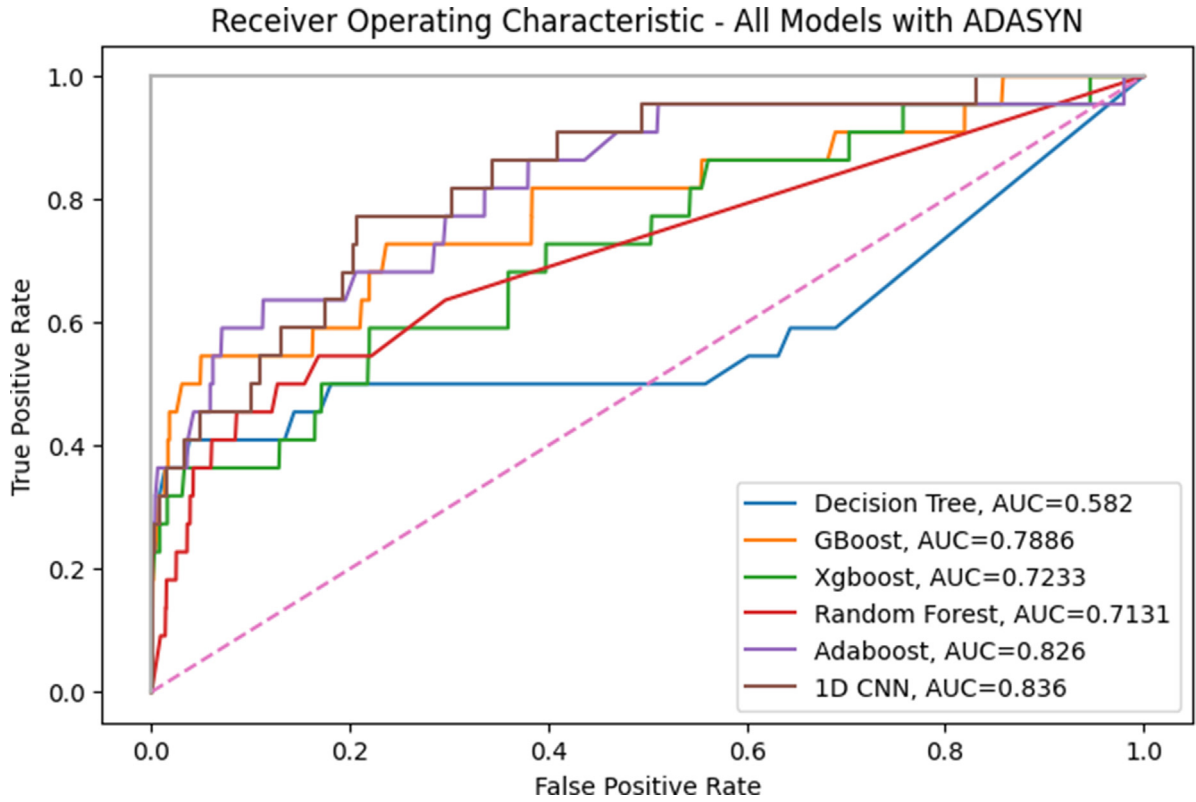


Fig. 8. Comparison of ROC-AUC values.

challenging, as models often bias toward the majority class (benign), leading to low true positive rates (Table 5). To address this, we first analyzed the impact of data balancing techniques in Section 4.3, demonstrating how resampling strategies mitigate class imbalance and enhance model performance.

In Section 4.4, our results show that the 1D CNN not only improves accuracy (outperforming existing methods, as shown in Table 8) but also significantly reduces the false positive rate (FPR) to 0.0008, the lowest among comparable approaches. By combining data-balancing preprocessing with the 1D CNN's hierarchical feature learning, our framework achieves robust fraud detection while minimizing misclassification of legitimate transactions, a critical advancement for real-world Blockchain monitoring.

Nevertheless, both machine learning (ML) and deep learning (DL) techniques exhibit distinct impacts on the detection of fraudulent activity depending on the circumstances. Tree-based machine learning models prove to be effective in identifying fraudulent transactions, particularly in scenarios with a relatively low number of data instances—both benign and fraudulent transactions. In our previous research (Hasan et al., [18]), we delved into various tree-based ML techniques for classifying anomalous Bitcoin transactions. Given the capability of ML models to discern transactions in datasets with limited instances, we introduced a novel undersampling method named XGBCLUS, comparing its performance against state-of-the-art methods. Furthermore, we extracted anomaly rules from these tree-based models, aiming for enhanced human understanding. Thus, we posit that ML models demonstrate efficacy in detecting anomalies or fraud in blockchain transactions when data instances are limited, balanced through under-sampling techniques, and the interpretation of anomaly rules is deemed essential for human comprehension.

Conversely, deep learning models, particularly those based on convolutional neural networks (CNNs), prove to be effective in detecting fraud or anomalies in blockchain transactions, especially when dealing with a substantial volume of data instances. CNN-based models excel

at learning features from large datasets, providing a robust framework for fraud detection. Additionally, in the context of data balancing, over-sampling techniques demonstrate effectiveness, as deep learning models can adeptly glean insights from extensive data. This study specifically explores the application of the 1D CNN deep learning method for fraud detection in blockchain transactions, employing various over-sampling methods for effective data balancing. While utilizing the same dataset for both studies, our future work aims to expand the scope by collecting additional blockchain transaction data to develop an even more robust real-time fraud detection system, as shown in Figure 1, for blockchain transactions.

Since Machine learning and deep learning models are often considered as "Black Boxes", to make these models more understandable, we turn to Explainable AI using SHAP, as discussed in section 4.5. By analyzing SHAP values across all dataset instances, we identify the most crucial feature, which, in this case, is "total_btc" for both ML and DL models. This feature's significance is clear to human observers. SHAP's analysis emphasizes the importance of "total_btc" in identifying fraudulent Bitcoin transactions. SHAP offers a clear and interpretable way to comprehend how each feature influences the fraudulent transaction detection process. This aids data analysts and domain experts in recognizing patterns, connections, and potential features in the data that the model uses for its predictions. This transparency is particularly valuable in fields where explainability is essential, enabling stakeholders to validate the model's decisions and identify potential data or model issues.

Our study's findings have significant implications for improving both blockchain security and anomaly detection. The effectiveness of the proposed CNN model in identifying anomalous Bitcoin transactions highlights a promising approach for enhancing blockchain system security. Additionally, incorporating eXplainable Artificial Intelligence (XAI) techniques adds transparency and clarity to the anomaly detection process. These explainability measures contribute to the development of robust anomaly detection systems, thereby strengthening the overall security of blockchain technologies. Furthermore, our exploration of various

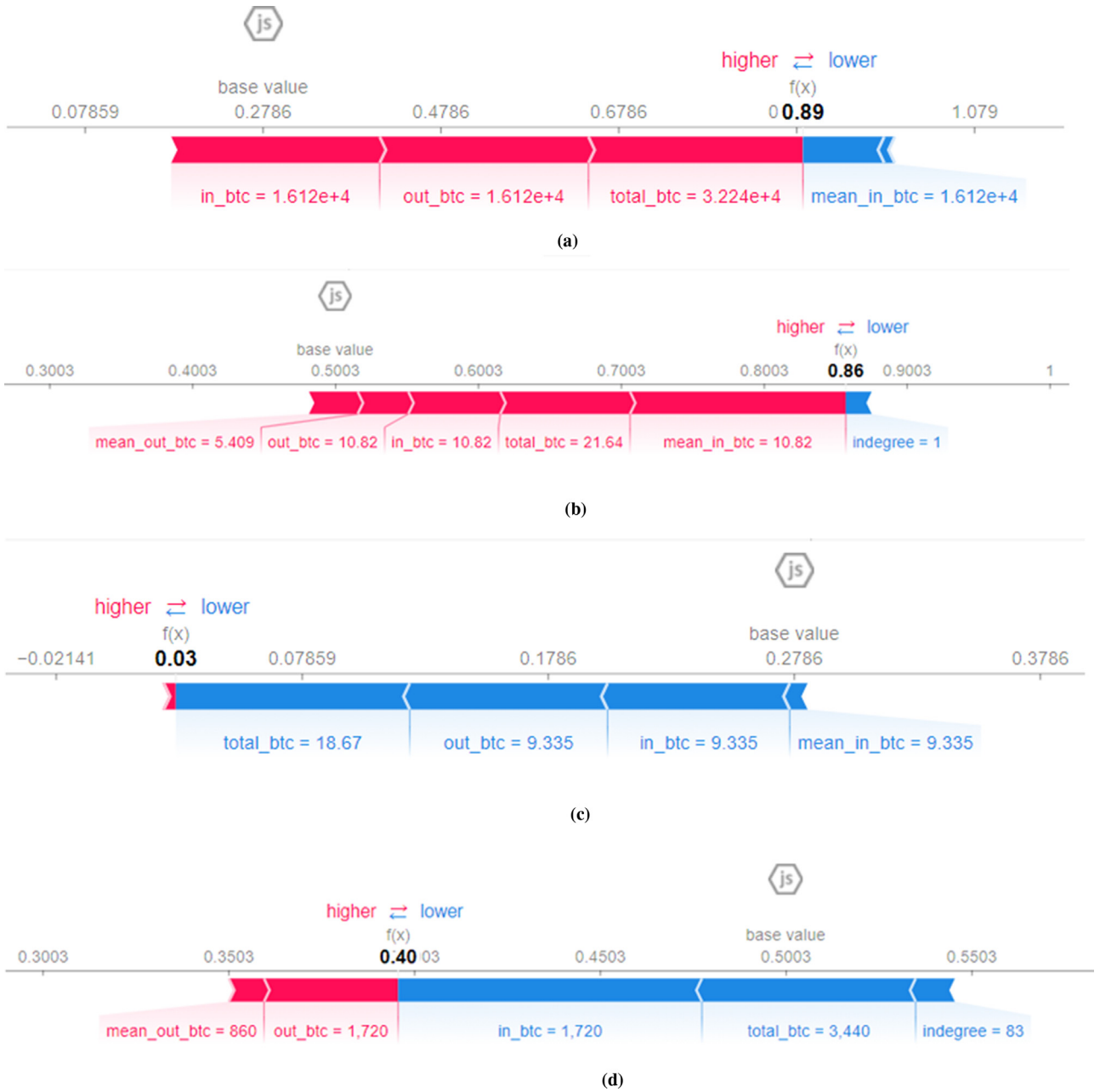


Fig. 9. Features contributing to (a) detect a fraudulent transaction with CNN model, (b) detect a fraudulent transaction with ML model, (c) detect a benign transaction with CNN, (d) detect a benign transaction with ML model.

sampling techniques provides valuable strategies for managing highly imbalanced datasets within the blockchain domain. This has broader applications for anomaly detection in other areas, such as credit card fraud detection and money laundering, where imbalanced data is a common challenge. Implementing these strategies can significantly improve detection accuracy and overall system reliability in various contexts.

Given the success of our deep learning approach in fraud detection, future research could extend the use of our CNN technique to other Blockchain domains like Ethereum transactions, credit card fraud detection, and money laundering. Investigating data-driven methods might lead to the creation of real-time Blockchain fraud detection systems. Additionally, combining machine learning and deep learning algorithms, such as CNN + SVM, could be explored to identify the better model for digital transaction anomaly detection.

6. Conclusion

In this paper, we have presented a 1D CNN model designed to identify fraudulent Bitcoin transactions. We also investigated various over-sampling and combined sampling techniques to address the significant class imbalance in Bitcoin transaction data, offering insights into effective balancing methods through a comprehensive comparative analysis. Further, we enhanced the interpretability of both CNN and ML classifier predictions by employing SHAP to identify the key features influencing Bitcoin transaction classification. Our rigorous evaluation proves the effectiveness of our proposed 1D CNN model in detecting fraudulent and benign Bitcoin transactions. Our proposed 1D CNN model achieves an accuracy of 0.99 and minimizing the FPR score to zero (0). In conclusion, our study demonstrates that the proposed CNN method, combined

with appropriate balancing techniques, is highly effective in detecting fraudulent Blockchain transactions.

Declaration of competing interest

The authors declare no conflict of interests.

CRediT authorship contribution statement

Mohammad Hasan: Conceptualization, Data curation, Methodology, Software, Writing – original draft. **Mohammad Shahriar Rahman:** Writing – review & editing. **Mohammad Javed Morshed Chowdhury:** Writing – review & editing. **Iqbal H. Sarker:** Conceptualization, Formal analysis, Supervision, Writing – review & editing.

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